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# Improving Marketing/ Logistics Cross-Functional Collaboration in the Supply Chain

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*Currently, little is known about the internal (intrafirm) behaviors that may positively affect collaborative marketing/logistics integration, or about the benefits that may be associated with making these considerable investments. In short, what should firms be doing to promote marketing/logistics collaborative integration? What is the potential payoff? This paper addresses these questions by examining relationships between the organization's evaluation and reward system, cross-functional collaboration, effective marketing/logistics interdepartmental integration, and distribution service performance. © 2000 Elsevier Science Inc. All rights reserved.*

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## INTRODUCTION

To balance customers' requirements with the need for profitable growth, many firms are aggressively focusing on improving supply chain management (SCM). The supply chain involves "all activities associated with the flow and transformation of goods from the raw material stage, through to the end user, as well as the associated information flows" [1]. SCM is the integration of these activities through improved inter- and intrafirm relationships to achieve sustainable competitive advantage.

It has been suggested that success in today's competitive business environment is largely dependent on the degree to which firms are able to integrate across traditional functional boundaries to provide better customer service

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[2–4]. With customers increasingly becoming more demanding, firms place more emphasis on customer service. Achieving better levels of customer service requires working together across departments or functions.

Providing customer service in the supply chain is largely the domain of two functional areas—marketing and logistics. Collaborative integration between a firm’s marketing and logistics functions is necessary to fully capitalize on potential service improvements [5–7]. However, such collaborative marketing/logistics integration may require the expenditure of substantial managerial and financial resources. Frequently, radical changes to traditional working practices are required as well.

Currently, little is known about the internal (intrafirm) behaviors that may positively affect collaborative marketing/logistics integration, or about the benefits that may be associated with making these considerable investments. In short, what should firms be doing to promote marketing/logistics collaborative integration? What is the potential payoff? This paper addresses those questions by examining relationships between the organization’s evaluation and reward system, cross-functional collaboration, effective marketing/logistics interdepartmental integration, and distribution service performance.

## BACKGROUND

It is widely agreed that task interdependence is the catalyst for interdepartmental integration. In simpler terms, customer satisfaction is dependent on the output of more than one worker, or one functional area. Numerous empirical studies suggest that collaborative cross-functional integration is positively associated with performance [8–10]. Although cross-functional integration is an important issue, most of the research to date has focused on marketing/research and development (R&D) integration and the

benefits associated with such integration. Empirical studies are needed to examine the potential for marketing/logistics integration and its impact on performance.

Effective marketing/R&D interdepartmental integration is still relatively rare in organizations [11, 12]. This also appears to be the case with marketing/logistics cross-functional integration. The limited studies that were identified indicate that marketing and logistics managers have tended not to consult and coordinate with each other [13–16]. In fact, marketing/logistics interdepartmental relations tend to be characterized by conflict and lack of communication rather than by collaborative integration. Research is needed to examine the potential contribution if marketing/logistics relationships are characterized by more collaborative interactions.

## Collaborative Cross-Functional Integration

Collaborative interdepartmental integration involves predominantly informal processes based on trust, mutual respect and information sharing, the joint ownership of decisions, and collective responsibility for outcomes [8, 9]. In short, collaborative integration is how well departments work together when their jobs require them to do so. Thus, collaboration between departments is often needed to ensure delivery of high quality services to customers, and involves the ability to work seamlessly across the “silos that have characterized organizational structures” [17]. Collaborative behavior is based on cooperation (willingness), rather than on compliance (requirement). Its success is contingent upon the ability of individuals from interdependent departments to build meaningful relationships [18, 19]. The fundamental challenge for managers focusing on improving customer service in the supply chain is to gain a better understanding of the antecedents and consequences of cross-functional collaboration.

## Organizational Evaluation and Reward Systems, Cross-Functional Collaboration, and Effective Interdepartmental Relations

Evaluation and reward systems are mechanisms that a firm can use to stimulate or foster cooperation between functional areas. Adequate incentives can bring together disparate individuals to achieve common goals [20]. Conversely, short-sighted evaluation and reward systems can create disincentives for collaboration.

Firms must ensure that their evaluation and reward systems are aligned with their business strategies. If

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# Collaborative interdepartmental integration is based on trust.

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functions are very interdependent in their work, it is counterproductive to base evaluation and reward systems on individual performance. The nature of such work demands compatible systems such as team-based pay and compensation, performance appraisal and accountability at the team level, and recognition for team results [21]. To promote interdepartmental collaboration, marketing and logistics personnel must be encouraged to think proactively about processes rather than discrete job functions. Thus, firms whose evaluation and reward systems recognize cooperation and teamwork may experience higher levels of marketing/logistics cross-functional collaboration and more effective marketing/logistics interdepartmental relations.

Effectiveness of interdepartmental relations involves the perceptions of personnel who interact with people in another functional area, that their relationship is worthwhile, equitable, productive, and satisfying [22, 23]. The effectiveness construct is often used to confirm that managers' attitudinal perceptions of behavioral initiatives are as expected. For example, in the current context, logistics managers reporting high levels of cross-functional collaboration are also expected to perceive that their departments' relationships with marketing are relatively productive and worthwhile.

Cross-functional integration is also important because it affects cycle time reduction, perceptions of customer value, and customer service [24]. Therefore, it is also anticipated that cross-functional collaboration and effective interdepartmental relations will be associated with better distribution service performance.

## **Cross-Functional Collaboration, Effective Interdepartmental Relations, and Distribution Service Performance**

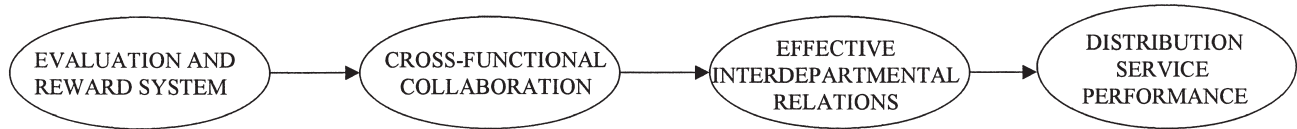
As more firms have discovered the significant opportunities for differentiation presented by managing the flow of product to the customer in better, more efficient ways, much attention has been focused on the measurement of distribution service performance. For example,

the ability to meet quoted or anticipated delivery dates and quantities on a consistent basis, the ability to respond to the needs and wants of key customers, and the ability to notify customers in advance of delivery delays or product shortages are identified as key distribution service performance areas [25].

Many successful firms are focusing on distribution competency to build relationships with key customers by customizing their basic service offering until it is "just different enough to fit exactly what the customer needs" [26]. The proliferation of quick response systems (QRs), efficient consumer response (ECR) initiatives, and just-in-time (JIT) supply programs are tangible examples of how a distribution service can offer customers added value. These programs tend to position a distribution service as the core capability that achieves customer satisfaction through inventory availability, timely delivery, lower product failure rates, and thus fewer lost sales or returns/complaints [27].

Successful distribution service performance is often dependent on the level of collaboration that exists between the firm's marketing and logistics functions. The absence of cross-functional collaboration may result in promises made by the firm's sales force that have not been coordinated with logistics, promotions that are not synchronized with delivery schedules, and failure to deliver product in a specific, requested format because it is not the most efficient way to do so. Without marketing/logistics cross-functional collaboration, firms cannot be expected to respond optimally to customers' requirements. It is, therefore, expected that effective interdepartmental relations—the product of cross-functional collaboration—will be positively associated with distribution service performance.

In summary, this study proposes that cross-functional collaboration and effective interdepartmental relations are influenced by the organization's evaluation and reward system, and that cross-functional collaboration and effective interdepartmental relations are positively associated with distribution service performance. The proposed relationships are summarized in Figure 1.



**FIGURE 1** Proposed relationships.

## METHODS

### Sample Design

The sampling frame consisted of logistics managers employed by U.S.-based manufacturers listed in the Council of Logistics Management (CLM) membership roster, where 2,046 potential candidates were identified. A mail survey methodology was employed.

A preliminary survey instrument was pretested with 12 logistics managers and educators who were asked to comment on the wording, presentation, and face validity of items in the instrument. Suggestions for the rewording and repositioning of items were incorporated into the final instrument. Next, researchers telephoned 360 logistics managers randomly drawn from the sampling frame, to ask for permission to mail out the questionnaire. One hundred and fifty-two logistics managers that were originally selected had to be replaced (17 had moved on; 22 were out of the office for the duration of the calling period; 7 no longer worked as logistics managers; 20 declined the opportunity to participate; and 86 had not been reached after 6 call attempts). The pre-notification process continued for 10 days, until 360 managers agreed to participate.

The initial mailing was comprised of a personalized request on university letterhead. In addition, a \$2 bill was included with each questionnaire (the inclusion was not mentioned in the prior telephone conversations). To further facilitate the process, return postage was also provided. Two hundred and sixty-one managers responded to the first mailing. An additional 48 managers responded to a second mailing that was sent to the 99 managers who had not responded after 4 weeks—for a total of 309 responses (60.4%). Table 1 provides descriptive statistics of the sample.

Non-response bias was assessed using procedures recommended by Armstrong and Overton [28]. The last one-quarter of responses received were assumed to be most similar to non-respondents, since their replies took the longest time and the most effort to obtain. Therefore, the last quartile was compared to the first three quartiles.

The comparisons of group mean responses to survey questions revealed no significant differences on the 20 variables analyzed, suggesting that non-response bias is unlikely to be an issue in interpreting the results of this study.

### Measure Development

Measures were selected based on past research and, where necessary, were adapted to fit the current context. Items from Barclay's study were utilized to assess managerial perceptions of organizational evaluation and reward systems [29]. Cross-functional collaboration was examined with a series of items adapted from Kahn's research that examined the impact of R&D, marketing, and manufacturing cross-functional integration on performance [9]. The items for perceived effectiveness of interdepartmental relations were adapted from Van de Ven and Ferry's work on the evaluation and assessment of organizations that has been employed in numerous more recent studies [30]. Finally, distribution service perfor-

**TABLE 1**  
Sample Characteristics (309 Logistics Managers)

| Annual divisional sales for the last financial year |                     |
|-----------------------------------------------------|---------------------|
| \$100 million or less                               | 10%                 |
| Over \$100 million to \$500 million                 | 37%                 |
| Over \$500 million to \$1 billion                   | 20%                 |
| Over \$1 billion                                    | 33%                 |
| Number of full-time employees                       |                     |
| 250 or less                                         | 16%                 |
| 251 to 500                                          | 17%                 |
| 501 to 1,000                                        | 16%                 |
| 1,001 to 2,000                                      | 19%                 |
| Over 2,000                                          | 32%                 |
| Industry description—leading categories             |                     |
| Food                                                | 52 managers (16.8%) |
| Consumer goods (non-food)                           | 44 managers (14.2%) |
| Chemicals                                           | 43 managers (13.9%) |
| Computers/communications                            | 36 managers (11.6%) |
| Pharmaceuticals                                     | 24 managers (7.8%)  |
| Automotive                                          | 19 managers (6.1%)  |
| Paper/packaging                                     | 16 managers (5.2%)  |

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# Short-sighted evaluation and reward systems can create disincentives for collaboration.

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mance was measured using items from the Michigan State Global Logistics research study [25]. All scale measures had Cronbach alpha values equal to or greater than 0.70 and were considered reliable according to Nunnally's guidelines for reliability [31]. The individual items and their means, as well as the coefficient alphas and overall mean scores for each scale, are shown in Table 2.

## ANALYSIS AND RESULTS

First, the mean scores for the scales, and for the individual items in each scale, are presented in order to evaluate current perceptions of organizational reward and evaluation systems, marketing/logistics cross-functional collaboration, the effectiveness of interdepartmental relations, and distribution service performance. Following this, the tests for the proposed relationships are described, and the results are presented and discussed.

### Evaluation of Mean Scores

Logistics managers were asked to evaluate several statements about their firms' reward and evaluation systems (based on a 7-point scale: 1 = strongly disagree; 7 = strongly agree). The overall mean score for the scale was 4.67, suggesting that logistics managers have only moderately favorable perceptions of their firms' evaluation and reward systems. Most significantly, respondents somewhat disagreed with the statement that "evidence of cooperation between departments is acknowledged by superiors in this organization" (3.35 based on a 7-point scale).

The relative lack of recognition for the advisability of interdepartmental cooperation is borne out by low mean scores for the cross-functional collaboration and the effectiveness of interdepartmental relation scales. Managers were asked to indicate to what degree the logistics department had pursued various collaborative activities

with the marketing department over the last six months (based on a 5-point scale; 1 = never; 5 = quite frequently). The overall mean score for the items in the scale that measured cross-functional collaboration was 2.96. The most frequently pursued collaborative activity is "informally working together" (3.47); the least pursued collaborative activities are "making joint decisions about ways to improve overall cost efficiency" (2.57), "developing a mutual understanding" (2.69), and "achieving goals collectively" (2.76). These findings suggest that marketing/logistics collaborative behavior occurs occasionally rather than often.

Respondents were also asked to report on the effectiveness of marketing/logistics interdepartmental relations. Not surprisingly, logistics managers' perceptions of their departments' relationships with marketing were only marginally higher than their perceptions of the frequency of interdepartmental collaborative behavior. The overall mean for the effectiveness scale was 3.24 (based on a 5-point scale: 1 = to no extent; 5 = to great extent). The highest mean score (3.67 based on a 5-point scale) was generated in response to the following item: "To what extent is the time and effort spent in developing and maintaining a relationship with marketing worthwhile." The lowest mean score (2.92) was generated in response to the following item: "To what extent has marketing carried out its responsibilities and commitments in regard to logistics."

These findings indicate that while logistics managers feel that the time and effort spent cultivating the relationship with marketing is worthwhile, interdepartmental relations are not particularly effective. They recognize the value of interdepartmental collaborative behavior, but there is little "follow-through" on a daily/operating basis.

Internal relationship effectiveness must also translate to externally recognized value. Thus, respondents were asked to evaluate how well their firms performed logis-

**TABLE 2**  
**Items, Coefficient Alphas and Mean Scores**

|                                                                                                                                                                                       |                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Reward (alpha .82)                                                                                                                                                                    | Construct mean 4.67 |
| How strongly do you agree or disagree with each of the following statements about the reward system at your company? (7-point scale: 1 = strongly disagree, 7 = strongly agree)       |                     |
|                                                                                                                                                                                       | Item mean           |
| My organization blames departments for errors rather than seeking the causes of errors.*                                                                                              | 2.57                |
| There is little recognition given for considering another department's problems.*                                                                                                     | 2.96                |
| People pretty well look out for their own interests.*                                                                                                                                 | 3.14                |
| Evidence of cooperation between departments is acknowledged by superiors in this organization.                                                                                        | 3.35                |
| Collaboration (alpha .93)                                                                                                                                                             | Construct mean 2.96 |
| During the past six months, how often did the logistics department engage in the following activities with the marketing department? (5-point scale: 1 = never; 5 = quite frequently) |                     |
|                                                                                                                                                                                       | Item mean           |
| Informally working together                                                                                                                                                           | 3.47                |
| Sharing ideas, information, and/or resources                                                                                                                                          | 3.19                |
| Working together as a team                                                                                                                                                            | 3.13                |
| Conducting joint planning to anticipate and resolve operational problems                                                                                                              | 2.92                |
| Achieving goals collectively                                                                                                                                                          | 2.76                |
| Developing a mutual understanding of responsibilities                                                                                                                                 | 2.69                |
| Making joint decisions about ways to improve overall cost efficiency                                                                                                                  | 2.57                |
| Effectiveness of interdepartmental relations (alpha .89)                                                                                                                              | Construct mean 3.24 |
| During the past six months, to what extent...(5-point scale: 1 = to no extent; 5 = to a great extent)                                                                                 |                     |
|                                                                                                                                                                                       | Item mean           |
| Has the time and effort spent in developing and maintaining the relationship with marketing worthwhile?                                                                               | 3.67                |
| has the relationship between logistics and marketing been productive?                                                                                                                 | 3.43                |
| have you been satisfied with the overall relationship between logistics and marketing?                                                                                                | 2.94                |
| has marketing carried out its responsibilities and commitments in regard to logistics?                                                                                                | 2.92                |
| Distribution service performance (alpha .79)                                                                                                                                          | Construct mean 3.74 |
| Relative to your division's largest competitor, how well do you perform in these areas? (5 point scale: 1 = much better; 5 = much worse)                                              |                     |
|                                                                                                                                                                                       | Item mean           |
| Responding to the needs and wants of key customers                                                                                                                                    | 4.02                |
| Accommodating special customer service requests                                                                                                                                       | 3.90                |
| Meeting quoted or anticipated delivery dates on a consistent basis                                                                                                                    | 3.75                |
| Providing desired quantities on a consistent basis                                                                                                                                    | 3.74                |
| Notifying customers in advance of delivery delays or product shortages                                                                                                                | 3.27                |

\*Indicates reverse-scored items (item mean scores have been reversed for overall construct mean).

tics activities relative to their firms' largest competitor (based on a 5-point scale: 1 = much worse; 5 = much better). The overall mean for the scale items was 3.74. Firms do best at "providing desired quantities on a consistent basis" (4.02) and worst at "notifying customers in advance of delivery delays or shortages" (3.27). The mean scores suggest that there are significant opportunities for firms to differentiate themselves by improving distribution service performance. This is consistent with

business practices at many firms today. Most firms are good at the basics, i.e., getting the product there. However, communication is often cited as a neglected area [26].

Raising the low levels of marketing/logistics cross-functional collaboration that are indicated by these findings may be a way to improve distribution service performance. Therefore, to gain a better understanding of cross-functional collaboration, the current research assessed the relationships outlined in Figure 1.

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# Distribution competency builds relationships with key customers.

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## Tests of Proposed Relationships

Pearson correlations were calculated to gain initial insight into bivariate relationships among organizational evaluation and reward systems, cross-functional collaboration, effective interdepartmental relations, and distribution service performance. Main effects of organizational evaluation and reward systems on cross-functional collaboration, of cross-functional collaboration on effective interdepartmental relations, and of effective interdepartmental relations on distribution service performance were estimated using simple linear regression analyses. Evaluation of the results is based on the overall robustness and fit of the regression models, and the direction and significance of individual parameters. Interactions were evaluated using the hierarchical multiple regression procedure proposed by Baron and Kenny to determine whether the flow of the relationships among the variables is linear and sequential as depicted in Figure 1 [32].

## Results and Discussion of Simple Linear Regression Analyses

The significant positive correlations for the following relationships all give preliminary evidence—on a bivariate basis—that the variables are related: between the organization's evaluation and reward system and its cross-functional collaboration; between cross-functional collaboration and effective interdepartmental relations; and between effective interdepartmental relations and distribution service performance. Correlations and results of the simple linear regression analyses conducted to estimate the strengths and directions of the relationships are reported in Table 3.

Preliminary examination of the main effects provided strong support for the proposed sequential links between the four constructs. Significant and positive beta coefficients, as well as highly significant fit statistics and explained variance, were found for the organizational evaluation and reward system/cross-functional collaboration,

cross-functional collaboration/effective interdepartmental relations, and effective interdepartmental relations/distribution service performance relationships. As shown in Table 3, the  $R^2$ 's for the three simple linear regression analyses were .131, .551, and .071, respectively.

Thus, 13.1% of the variance in cross-functional collaboration can be attributed to the organization's evaluation and reward system. In simpler terms, this result suggests that correctly aligning a firm's evaluation and reward system can have a major impact on cross-functional collaboration. When a reward system encourages employees to aim for broader goals rather than myopically focusing on meeting departmental objectives, the employees may identify more with the organization and may be more prepared to collaborate with other departments.

Similarly, 55.1% of the variance in effective interdepartmental relations is explained by cross-functional collaboration. In other words, the presence or absence of cross-functional collaborative behavior appears to largely determine whether interdepartmental relations are perceived to be effective. Managers feel far more satisfied with the overall relationship between marketing and logistics when collaborative behaviors occur frequently.

The most important association from a managerial perspective is to assess the benefits that may result from improving cross-functional collaborative behavior and perceptions of effective interdepartmental relations. Therefore, it is particularly significant that the results indicate that effective marketing/logistics interdepartmental relations explain 7.1% of the variance in distribution service performance. This finding provides empirical evidence that the effectiveness of interdepartmental relations positively impacts distribution service performance. Research indicates that firms must routinely offer customers high levels of basic distribution service performance to ensure continued business [26]. Accordingly, the potential of a 7% improvement in distribution service performance is substantial and may be enough to significantly differentiate a firm.

# The time and effort spent cultivating the relationship with marketing is worthwhile.

## Results and Discussion of Hierarchical Regression Analyses

The simple linear regression analyses reveal a sequence of discrete associations for the relationships outlined in Figure 1. To further understand the sequencing and relationships, hierarchical regression analyses were performed. Estimations based on hierarchical regression analyses are presented in Table 4.

Analysis 1 in Table 4 examines the linearity of the organization's evaluation and reward system, cross-functional collaboration, and effective interdepartmental relations. The initial findings suggest a significant direct association between the organization's evaluation and reward system and its effective interdepartmental relations. However, the addition of collaboration increases the re-

gression model's  $R^2$  from .178 to .561. Thus, the sequential impacts of the organization's evaluation and reward system and its collaboration explain more than 56% of the variance in effective interdepartmental relations, while the organization's evaluation and reward system on its own only explains 17.8% of the variance in effective interdepartmental relations.

The significant incremental  $R^2$  of .383 indicates that collaboration is a necessary mediator of the relationship between the organization's evaluation and reward system and its effective interdepartmental relations [32]. In short, although a direct significant association exists between the organization's evaluation and reward system and effective interdepartmental relations, the inclusion of collaboration helps to explain the influence of the organization's evalua-

**TABLE 3**  
Correlations and Results of Simple Linear Regression Analyses

| Correlations            |                     |                     |                       |
|-------------------------|---------------------|---------------------|-----------------------|
|                         | Reward              | Collaboration       | Effective relations   |
| Reward                  | —                   | —                   | —                     |
| Collaboration           | .382 <sup>a</sup>   | —                   | —                     |
| Effective relations     | .421 <sup>a</sup>   | .732 <sup>a</sup>   | —                     |
| Logistics performance   | .214 <sup>a</sup>   | .250 <sup>a</sup>   | .266*                 |
| Regression Analyses     |                     |                     |                       |
| Independent variables   | Dependent variables |                     |                       |
|                         | Collaboration       | Effective relations | Logistics performance |
| Reward                  | .362<br>(.000)      | —                   | —                     |
| Collaboration           | —                   | .742<br>(.000)      | —                     |
| Effective relations     | —                   | —                   | .266<br>(.000)        |
| Model F                 | 45.035              | 364.622             | 21.555                |
| (p-value)               | (.000)              | (.000)              | (.000)                |
| Adjusted R <sup>2</sup> | .131                | .551                | .071                  |

\*Coefficient sig at  $p < .01$ .

Note: Beta coefficients are reported with p-values based on t-statistics shown in parentheses. The variables listed in the first column show the independent variables included in the regression analyses. The variable at the head of each remaining column shows the dependent measure. All estimations are based on 309 observations.

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# The overall goal of providing superior customer service may be jeopardized by a shortage of cross-functional collaboration.

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tion and reward system on effective interdepartmental relations far more accurately and meaningfully.

Analysis 2 in Table 4 examines the proposed linearity of collaboration, effective interdepartmental relations, and distribution service performance. The initial results suggest a significant positive association between collaboration and distribution service performance. However, further analysis shows that when effective interdepartmental relations is added to the model, the direct relationship between collaboration and distribution service is no longer significant. In addition, the inclusion of effective interdepartmental relations increases the regression model's  $R^2$  from .065 to .080 (an  $R^2$  increment of .015).

These findings suggest that effective interdepartmental relations mediate the relationship between collaboration and distribution service performance and, therefore, must be included in the analysis for a more accurate and meaningful representation of the sequence of relationships between the three constructs. In other words, the true relationship between collaboration and distribution service performance is only accurately represented when effective interdepartmental relations is included as a mediating influence.

As stated earlier, the main objective of this paper is to gain a better understanding of the antecedents and consequences of marketing/logistics cross-functional collaboration to assess benefits that may be associated with encouraging such integrative behavior. Therefore, Analysis 3 in Table 4 tests the overall robustness of the sequence of relationships proposed in Figure 1. The findings show progressively more significant  $R^2$  statistics as each element of the model is added. On its own, the organization's evaluation and reward system explains 5.5% of the variance in distribution service performance. However, when collaboration is added, 8.9% of the variance in distribution service performance is explained (an  $R^2$  increment of .034).

The complete series of associations proposed in Figure 1 yields an  $R^2$  statistic of .096, indicating that nearly 10% of the variance in distribution service performance can be explained by the combined sequential impact of the organization's evaluation and reward system, cross-functional collaboration, and effective interdepartmental relations. Considering the multiple resources and environmental factors that also affect distribution service performance, it is significant that logistics managers perceive collaboration with their marketing counterparts to have so great a potential influence. This finding has significant managerial implications.

## MANAGERIAL IMPLICATIONS

In SCM, individual tasks or functions must often be subordinated to achieve overall goals. For example, marketing and logistics have joint responsibility for customer service. However, the results of this study indicate that marketing/logistics interdepartmental relations are only moderately effective. Accordingly, the overall goal of providing superior customer service may be jeopardized by a shortage of cross-functional collaboration.

When working relations between marketing and logistics are poor, the coordination and communication that is crucial for the provision of optimal distribution service may be lacking. In addition, personnel may divert considerable attention and effort from serving customers to internal issues like turf protection, and deflecting blame for errors and shortfalls. In contrast, our results suggest that collaborative marketing/logistics interdepartmental relations can help firms provide superior service systems by developing a mutual understanding of responsibilities, sharing ideas, information and resources, and working together as a team to resolve operational problems.

To improve customer service in the supply chain, firms should use evaluation and reward systems that pro-

**TABLE 4**  
**Results of Hierarchical Regression Analyses**

| Independent variables          | Dependent variables |                     |                   |                   |                   |                   |                   |
|--------------------------------|---------------------|---------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                                | Analysis 1          |                     | Analysis 2        |                   | Analysis 2        |                   |                   |
|                                | Effective relations | Effective relations | Dist. serv. perf. | Dist. serv. perf. | Dist. serv. perf. | Dist. serv. perf. | Dist. serv. perf. |
| Evaluation and reward system   | .421<br>(.000)      | .170<br>(.000)      | —                 | —                 | —                 | —                 | —                 |
| Cross-functional collaboration | —                   | .669<br>(.000)      | —                 | —                 | —                 | —                 | —                 |
| Cross-functional collaboration | —                   | —                   | .255<br>(.000)    | .123<br>(.151)    | —                 | —                 | —                 |
| Effective relations            | —                   | —                   | —                 | .179<br>(.036)    | —                 | —                 | —                 |
| Evaluation and reward system   | —                   | —                   | —                 | —                 | .235<br>(.000)    | .161<br>(0.10)    | .138<br>(.032)    |
| Cross-functional collaboration | —                   | —                   | —                 | —                 | —                 | .198<br>(.002)    | .110<br>(.200)    |
| Effective relations            | —                   | —                   | —                 | —                 | —                 | —                 | .131<br>(.137)    |
| Model F                        | 63.061              | 186.105             | 19.194            | 11.925            | 15.990            | 13.282            | 9.635             |
| (p-value)                      | (.000)              | (.000)              | (.000)            | (.000)            | (.000)            | (.000)            | (.000)            |
| R <sup>2</sup>                 | .178                | .561                | .065              | .080              | .055              | .089              | .096              |
| Incremental R <sup>2</sup>     | —                   | .383                | —                 | .015              | —                 | .034              | .007              |
| F                              | —                   | 254.421             | —                 | 4.419             | —                 | 10.044            | 2.224             |
| (p-value)                      | —                   | (.000)              | —                 | (.036)            | —                 | (.002)            | (.137)            |

Note: Beta coefficients resulting from hierarchical regression analyses are reported with p-values based on statistics shown in parentheses. In the first column are the independent variables included in the regression analyses. The variable listed at the head of each remaining column shows the dependent measure.

mote marketing/logistics cross-functional collaboration and effective interdepartmental relations. There is little empirical evidence about the relative effectiveness of evaluation and reward strategies for the firm's marketing and logistics functions. However, based on Coombs and Gomez-Mejia's prescriptions for achieving cross-functional integration in high technology firms, the following suggestions may help supply chain managers encourage marketing/logistics cross-functional collaboration [33]:

- Base evaluation and reward systems on customer service performance. By linking recognition and remuneration to marketing/logistics cross-functional collaboration, it is possible to improve the degree of cooperation and mutual problem solving across these functional areas. This will also encourage information sharing and a culture that supports joint endeavors.
- Explicitly incorporate cross-functional collaboration as a major factor in the performance appraisal of all personnel—from functional managers to rank-and-file em-

ployees—involved in the marketing/logistics interface. This should make it clear that individualistic behavior will be penalized and collaborative behavior will be rewarded.

- Use performance indicators that reflect the contribution of both functional areas. For example, customer service surveys, as well as feedback from senior management visits to key accounts, can be used as a basis for determining bonus compensation, rather than function-specific performance measures like the number of new accounts or logistics cost control.

As previously stated, managing marketing/logistics cross-functional collaboration in the supply chain is increasingly important to firms because of the growing need for service-based differentiation. In many industries—particularly those that involve commodity or convenience goods—firms have reached such a level of parity on product, price, and promotion that customer service is what differentiates one supplier from another. Therefore, the true value of marketing/logistics cross-functional col-

laboration lies in its potential to enhance the firm's performance in the marketplace.

The significant positive association between effective marketing/logistics interdepartmental relations and distribution service performance suggests that supply chain managers should consider the benefits that may accrue from cross-functional collaboration. It is often assumed that marketing and logistics functions work together in harmony. However, in reality these entities are often not natural allies. They do not always think alike or hold the same values. The firms that find out how to minimize the traditional separation and disparity that continue to characterize marketing/logistics interdepartmental relations may be the firms that have the most successful service differentiation strategies.

## CONCLUSIONS

The results of the current research suggested that there are performance benefits associated with encouraging effective marketing/logistics interdepartmental relations, and that an appropriate evaluation and reward system that recognizes teamwork and cooperation is a significant catalyst for the promotion of cross-functional collaboration. The positive link between interpersonal communication strategy and distribution service performance indicated by the results of this research should be of particular interest to supply chain managers and academics. Researchers have often suggested that the integration of interdependent functions improves performance. However, studies that empirically support this association are scarce—particularly in the marketing/logistics area.

As the supply chain perspective makes distribution service even more critical for long-term success, firms will need to develop a better understanding of marketing/logistics interdepartmental integration. Future research should further investigate the antecedents and consequences of marketing/logistics cross-functional collaboration. For example, managers' perceptions of their firms' interdepartmental integration could be linked to secondary financial measures of firm performance to further validate the findings presented in this study.

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